

GDSN 162: Ex.3: Linking: Part 2

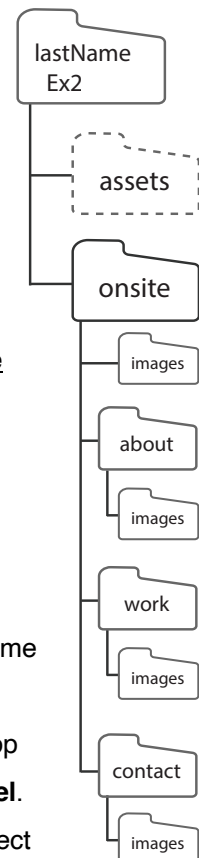
Objective:

Understand and explore image-based links. Learn intermediate linking methods such as:

Anchors, Hotspots/Image Maps, email links and Download Links and the terms associated with them. *As with all exercises, you should always take notes during the in-class demos and use these notes with the following instructions for a complete understanding of the assignment. Since Mac OS is the main platform used in our classrooms and in the design profession, the instructions are tailored for that platform. If you have a PC, it is easy to 'Google' the equivalent OS steps needed to perform the instructed shortcuts and actions.*

Process:

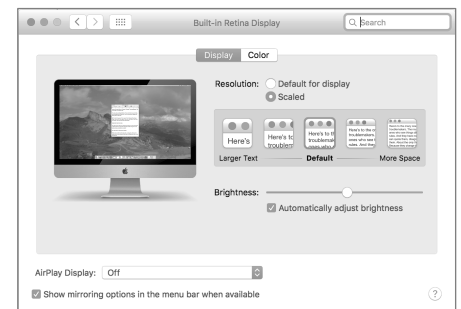
- 1) **Create a Site Folder, Assets Folder & Root Directory:** Create the same basic site you created in Exercise 2 (ex2). You could use the exact same site BUT you would not be practicing what you learned. Practice through repetition is what learning is all about. The more you work in **DW** (DreamWeaver) the more you will learn.
 - a) Create a folder and name it: **lastnameEx3** (for example, mine would be *gardosEx3*) this is your Site Folder.
 - b) Inside of this folder create a folder named **assets** (stuff not on the actual site) and a directory (stuff ON the actual site) named **onsite**.
This is your root directory.
 - c) In the **onsite directory**, set-up the same site structure as in ex2: You need an; **images directory, about directory, work directory and contact directory**.
Remember, an **images directory** need to be inside every directory in the onsite directory. If we think of the website as a house, and the sites' sections as the 'rooms' in our house, the images directory is the 'closet' for every room. This is where the images (for ex.; gifs, pngs and jpgs) are kept.
 - d) Create a sitemap which reflects this site structure (**lastnameEx3SiteMap.pdf**).
 - e) **Create identical pages and links that you created in ex.2.** The site structure should look exactly like the site you ended up with in ex.2. Same link names, same structure **except:**
 1. Go to: **Menu → File → Page Properties**. The Page Properties window will pop up. You can also click on the **Page Properties button** in the **Properties Panel**. Select **Appearance (CSS)** from the left. From the right, select **Page font**. Select the sans-serif type family list in the Appearance CSS setting that contains the



- typeface **Impact**. (we would never *actually* use Impact as it is difficult to read. We use it here both to show the legibility issue and to show a type change from Ex.2)
2. Change the CSS link colors of the Link, Visited, Rollover & Active to colors **different than what you used in ex.2**
 3. Use different Background colors than you used in ex.2 for the pages.
 4. Page titles need to be changed to: ***Lastname Home :: Ex3, Lastname About :: Ex3, Lastname Work :: Ex3, Lastname Contact :: Ex3***
 5. Link all the pages with the text-based links and assure everything is working by testing it in a browser.

Adobe Photoshop changes since Ver. 22:

- Please review the changes made to PSD since ver. 22
 - There are videos and instructions about these changes inside the **Course Resources and other Goodies! Module** of our Canvas site and on the **Exercise 1 page** of our Canvas site.
 - Particularly important to your workflow are the instructions on how to work with and configure:
 - The changed **Save As** option.
 - The default **Save to Cloud** option back to saving on your computer.
 - Getting rid of the **Welcome Screen** and the **New Document Window**.
- 2) Creating image-based links (PSD): Open Photoshop (PSD). Set-up the interface and close all unused panels (shown in our in-class lecture/demo and in the Canvas video) to make it easier to use. Go to: **Menu → File → New**. The New Document window will open.
- a) Name the document: ***lastnameEx3Nav.psd***.
 - b) Width: **120 pxls**
 - c) Height: **35 pxls**
 - d) Resolution: **72 Pixels/Inch (dpi)**
 - e) Color Mode: **RGB Color; 8 bit.**
 - f) Background Contents: **White**
 - g) Click **OK**. This will launch the new file.
 - h) We are working on high-resolution 5k monitors so this image may look really small. You can do a couple of things IF you want to see this image larger:
 - You can **Zoom-in** on the image: **Menu → View → Zoom in**. The image will be larger. It may look blurry or bit-mapped because you are making the pixels larger.



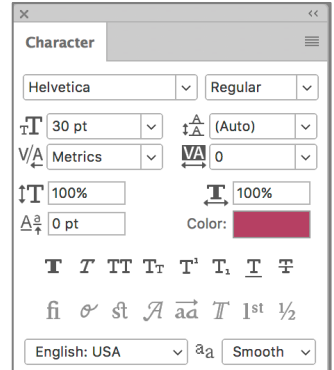
It will look fine on the web page.

- Or you can adjust the resolution of the display. Go to: **Menu → Apple → System**

Preferences → Displays. Click the Scaled button and select the larger text option.

- i) Select the **Type Tool** (the button with a capital T) from the **Toolbar**. Go to:

Menu → Window → Character to launch the **Character (Type) Panel**. You can use the **Control Bar** or the Character Panel to for your type settings.



1. Choose **Helvetica** (or the closest san-serif typeface you have) for the typeface. Set the font size to 30 points. Click the **Left align text button** in the Control Bar. Choose a dark color. Type the word **Home**. If you look at the **Layers panel**, **Home** is an editable type layer at the top of the panel.
2. In the **Layers panel**, click and drag the **Home layer** to the **Create a new layer** icon at the bottom of the layers panel. You should now have a layer called: **Home copy**. Turn off the visibility for the original **Home** layer by clicking the eye icon. ****Note: The text may look blurry or pixelated if you are zoomed-in. This is normal.**
Go to: Menu → View → Select 100% to see the file at a normal size.

3. Select the **Home copy layer**. Select the **Type Tool** and select the word **Home** on the *lastnameEx3Nav* image by clicking and dragging on the type. Change **Home** to the word **About**. Drag the **About layer** to the **Create New Layer** icon at the bottom of the layers panel. Turn off the visibility for the original **About layer**. With the **Type Tool** select the word **About** on the image and change it to **Work**.

	T	Home
	T	About
	T	Work
	T	Contact
		Background

4. Repeat the same steps to create a layer with the word **Contact**.
 5. You should now have a **Home layer**, **About layer**, **Work layer**, **Contact layer** and a **Background layer** (default).
- j) If needed, click and drag the layer order so **Home** is at the top, then **About**, **Work**, **Contact**. **Background** need to be at the bottom.



1. Select the **Move Tool** from the **Toolbar** (upper left, 4 intersecting arrows). Make all the layers visible. **Command + Click** and select all the layers **except the background**.
2. Align all the words. At the top of the PSD interface, on the **Control Bar**, click **Align Horizontal Centers** (icon with 2 thick horizontal bars and a thin vertical line in the center). Use the **Move tool** to position all the selected layers so they are in the approximate center of the white background. Save your work. **File → Save**.



3. Since this is a PSD file, and will not be used directly on your site's pages, save the file inside your Assets folder which is inside your *lastnameEx3* folder. The file needs to have the name **lastnameEx3Nav.psd** that you typed in when you created the new file. **Make sure the format at the bottom of the Save window is set to Photoshop**. Click

OK when the Maximize compatibility window pops up. This assures you can open the file with older versions of the software. Don't close the file.

- k) With the file still open, **turn off the visibility** for all layers **except** the **Background** and **Home** layer. Go to: **File → Export → Save for Web (Legacy)** and save as a .png file named **homeNav.png** inside the **images** directory inside your **onsite** directory (also called your root directory). **Do not change the name or the upper/lower case. Label it EXACTLY as the brief directs. Do not save these navigation images inside of any of the sub-directories (about, work, contact).**
- l) Repeat this step for the other 3 navigation images (**aboutNav.png, workNav.png, contactNav.png**) saving them all into the same top-level images directory inside the root directory. The targeted PSD layer doesn't have to be selected to save the image. Only the visible content will be saved as an optimized image. Save and close the **lastnameEx3Nav.psd** file, it's done!
- 3) **Adding Images to Page Content:** If you haven't done this already, open DW and **Manage your site**. Make sure to choose the **onsite directory** inside the **lastnameEx3** folder. The assets folder **should not be visible** inside the **Files window** in DW.
- a) Make sure you have re-created the same site structure as ex.2 with the required changes mentioned in Step 1) before you proceed.
- b) From the **Files** panel, open **index.htm**. Look at the source code:
- ```
<body>Home | About | Work | Contact</body>
```
- c) On the top of the **visible page**, click and insert the cursor to the right of the text-based Contact link. Hit the **Return key** to create a paragraph <p> break.
- d) Look at the source code:
- ```
<body>
<p>Home | <a href="about/about.htm">About</a> | <a href="work/work.htm">Work</a> |
<a href="contact/contact.htm">Contact</a></p>
<p>&nbsp;</p>
</body>
```
- You now have: an opening <p> tag in front of **Home** and a closing </p> tag behind the Contacttag.
- You also have: a new paragraph with a non-breaking space placeholder below it:
- ```
<p> </p>
```
- before the closing body tag </body>. All that by just hitting return!
- e) Go to: the HTML tab of the **Insert panel** at the top of the DW workspace. Click on the image icon (2<sup>nd</sup> from the left). You can also go to **Menu → Insert → Image**.



- f) From the pop-up window, browse to the *images directory* inside your *root directory*. Select the **homeNav.png** image and click Open. The image will be inserted in the page under your text-based links.
- g) Click to the right of the **Home image** you just inserted and repeat the same steps inserting all four navigation images in a row (*homeNav.png*, *aboutNav.png*, *workNav.png*, *contactNav.png*). You can also drag and drop images from your **Files panel in DW** onto your page. NEVER DRAG AN IMAGE FROM YOUR COMPUTER DIRECTLY INTO DW!  
**Have a look at the code of the pages by clicking Split to get a feel for what the inserted image code looks like.**
- h) Repeat the same navigation image insertion on the other 3 pages in your site (*about.htm*, *work.htm*, *contact.htm*) using the same images from the *images directory* in your *root directory*. Look at the path structure. Some of the links will have the **../images path** which means go out of the current directory and into the main images directory in the root level.
- **You only need 1 set of navigation images.** Using the same image on multiple pages is possible because the single image creates an **'instance'** each time it is placed on a page. An instance is kind of like a visual clone of the image that is in the main images directory in the root. Images used repeatedly on multiple pages throughout the site (like the upper left logo, navigation images and background images), are known as **global assets**. You do not need a set of navigation images in any other directory aside from the set that is in the main *images* directory in your root!
- 4) **Linking Images:** From the DW **Files panel**, open *index.htm* or if it's open, click on the *index.htm* tab to select that page. Linking images is very similar to linking text.
- a) Select the *About* image on the page by clicking on it. We won't link to the Home button because we are already on the home page (*index.htm*). In the **Properties Inspector panel** at the bottom of your workspace you will see the **Link** field under **Src**.
- 
- Next to the **Link** field you will see a folder icon. Click the folder icon and browse inside of your root directory to find the **about.htm** file. Select the file and click **Open**. You should now see **about/about.htm** in the *Link* field. The *about* part tells the link that the page is in the *about* directory. The */* tells the link to go into that directory. This is known as the file path.
- b) Also in the *Properties Inspector* panel is an **Alt** field. This field is VERY important for designing accessible sites. The text you put here can be found by search engines AND it can be read aloud by a computer screen reader (like Siri) for users with disabilities. Although not a requirement for this assignment, you need to put some text here to get used to using it. Write something as simple as: **Link to the about section**
- c) Repeat the same steps for the **Work** and the **Contact** image links on the *index.htm* page.

- d) From the Home page (*index.htm*) There is no need to link the *Home* image link to the *Home* page, *you are already on that page*. The same applies to the following; *About* page (no link on the *About* image), *Work* page (no link on the *Work* image), *Contact* page (no link on the *Contact* image).
  - e) Go to the other 3 pages and link all the image links to their destination pages.
  - f) Make sure you link the images *to the .htm pages* and **not to the images**.
  - g) Go to **File → Save All**. Then go to: **File → Real-time Preview** and choose Firefox to test the links. Make sure everything is working. ***Have a look at the code of the pages by clicking Split to get a feel for what the image link code looks like.***
- 5) **Anchor Links**: Anchor links are useful when navigating large amounts of vertical content on a web page.
- a) From the Files panel, open the *work.htm* page. Insert your cursor to the right of the **Contact navigation image** and hit the **Return key** to create a paragraph `<p>` break. There should be a blinking cursor under your image links.
  - b) Type the following text: **Page Middle | Page Bottom**. After the word **Bottom**, hit the **Return key** 12 times to create 12 paragraph breaks: `<p>&nbsp;</p>`.  
You can count them in the code view.
  - c) At the 12<sup>th</sup> return, type the word **Middle**. Next, **Shift + Return** to insert a **Line Break**: `<br>`.  
You can also insert a Line Break by going to: **Menu → Insert → HTML → Character → Line Break**. Next, type (with the square brackets): **[top]** just underneath the word **Middle**.
  - d) Insert your cursor to the right of **[top]** and create 12 more `<p>` breaks by using the Return key. At the 12<sup>th</sup> return, type the word **Bottom**. **Shift + Return** to insert a **Line Break** `<br>`.  
Then type: **[top]** just underneath the word **Bottom**. Hit the Return key 3 more times to create 3 more `<p>` breaks.
  - e) Scroll back to the top of the page and **insert your cursor just before the H in the text-based nav link *Home*** at the very top-left of the page.
    - In the **Properties Inspector panel** at the bottom, with the **HTML** tab on the left selected, under the **Format** area you should see **ID** with the word *None*. Click in the **ID field** and type the word **top** (replacing the word *None*) and **hit Return**. You may see a cute **anchor icon** appear to the left of the text-based *Home* link on the top left of the page.
  - f) Scroll to the middle of the page and select the word **Middle**. In the **Properties Inspector panel**, enter the word **middle** in the **ID field** and **hit Return** to commit the change. Under the word *Middle*, select the word: **top** in between the square brackets. In the **Properties**



- Inspector panel**, in the **Link field**, type **#top** and **hit Return** to commit the change. The **#** is the anchor and the word **top** is its name.
- g) Scroll to the bottom of the page and select the word **Bottom**. In the **Properties Inspector panel**, enter the word **bottom** in the **ID field**. **Hit Return** to commit the change. Select the word: **top** in between the square brackets under the word **Bottom**. In the **Properties Inspector panel**, in the **Link field**, type **#top** (with the hashtag) and **hit Return** to commit the change.
  - h) Scroll to the top of the page, select the words: **Page Middle**. In the **Properties Inspector panel**, in the **Link field**, type **#middle**. **Hit Return** to commit the change. At the top of the page, select the words: **Page Bottom**. In the **Properties Inspector panel**, in the **Link field**, type **#bottom**. Hit the Return key to commit the change.
  - i) You now have a **top** anchor on the text-based **Home** link, a **Middle** anchor in the middle of the page and a **Bottom** anchor at the bottom of the page. You also have **top** links at these lower anchors so you can get back to the top. **File → Save All**. Preview the page in a browser, size the browser so it is short in height and test the results so you can see the links jump from anchor to anchor.
  - j) For extra credit, you can try and insert the **smoothScroll Javascript** demonstrated in class and in the video tutorial on the Canvas site. You will need to download the **162Ex3\_xc\_smoothScroll.zip**. Follow the directions on the **smoothScrollREADME.rtf** doc. ***This must work to get the extra credit***. Make sure your attempt does not break the requirement of standard same-page anchor linking. You may want to make a copy of the root and manage it before attempting the extra credit. ***Have a look at the code of the pages by clicking Split to get a feel for what the image link code looks like.***
- 6) **Email link:** An email link is essential to all websites. You should be able to copy and paste the address and/or launch the users email app. When you visit a site. You will create a **text-based email link** and find or create an **image-based email link** on your **Contact page** (**contact.htm**).
- a) Find or make an image to communicate an email link (envelope, mailbox etc.). The email image needs to be clear and obvious to your user. If you find one online, you may want to re-size it and save an optimized version. Approx. **128px X 128px (icon size)** or smaller is a good size to aim for but you can experiment. Just don't make it too large.
  - b) Save the optimized email image inside the *images directory* inside the *contact directory*. Since the contact.htm page is the *only place* this image will appear on the site, it goes in the *contact images directory*.
  - c) From the **Files panel**, open the **contact.htm** page. Insert your cursor to the right of the **Contact** navigation image and **hit the Return key** to create a paragraph <p> break.

- d) Insert a text-based email link by typing an email address in the proper format. For example: **info@riohondo.edu**. (this way, the user can copy-paste the address if they are using a device that their email is not hooked up to). The email address you type does not have to be real.
- Highlight the text email address and in the **Properties Inspector panel**, in the **Link field**, type the email code format: **mailto:** and then place your own email link. Example: **mailto:info@riohondo.edu**
- e) To the right of the *text-based email link*, create a single space with the spacebar and insert your *email image* from the *contact images directory*.
- Select the email image and in the **Link field** on the **Properties Inspector panel**, type the email code. Example: **mailto:info@riohondo.edu** hit the **Return key** to commit the change.
  - **File → Save All**. Preview the page in a browser. Clicking the text-based email link OR the image-based email link **need to launch the Mail program** on your computer. ***Have a look at the code of the pages by clicking Split to get a feel for what the image link code looks like.***
- 7) **Download Links:** File downloads are a large part of website interactivity. You will create a simple download link on your **About page** (*about.htm*) to the **162ex3Brief.pdf** available on the Canvas LMS site.
- a) Find an image to communicate a pdf file link. This needs to be an **Adobe Acrobat icon**. It needs to be clear to your user. You can download an existing image or make your own. It need to be small. **Approx. 128 X 128 pixels (icon size)** or smaller. You can resize it as needed in PSD. Optimize and save it in the proper format for the type of image it is inside the *about images directory*.
- b) Download a copy of the 162ex3Brief.pdf and place it in the **about directory** (***not*** the *about images directory*). It needs to be in the same place as the *about.htm* doc.
- c) From the **Files panel**, open the *about.htm* page. Insert your cursor to the right of the **Contact** navigation image and **hit the Return key** to create a paragraph <p> break.
- d) Inset a text-based pdf download link by typing: **Click here to download the PDF document (#k)**. You need to always indicate to your user how big the file is by putting the actual numeric size in 'k' where the hashtag is. For example: **(256k)**. Clicking on a file (like the **ex3Brief.pdf**) and using **Command + i** will bring up the info window where you can get the actual file size. The class demo will show other ways of getting the file size.
- e) Highlight the text-based pdf download link. In the **Properties Inspector panel**, in the **Link field**, click the folder icon. **Link the text to the actual PDF file** in the *about directory*. Set the **Target** to **\_blank** from the drop-down menu next to the **Target field**. This will launch



the PDF in a new window or may just download the file. Browsers are configured differently how they handle downloads.

- f) To the right of the *text-based PDF link* create a space with the spacebar, insert your **PDF icon image** from the *about images directory*. Select the image by clicking on it. In the **Properties Inspector panel**, in the **Link field**, click the folder icon and **link the PDF icon image to the actual PDF file in the *about directory***. Set the **Target** to **\_blank** from the drop-down menu next to the **Target field**. **File → Save All**. Preview the page in a browser.
- 8) **Hotspot/Image Map Links:** A Hotspot or Image Map allows you to create multiple links from a single image. You will create 3 organically-laid out links from a single image on your **Contact page** (*contact.htm*). These links need to overlap like on the example. **Your Image Map image must be original and have different links than the example!**
- a) In PSD, create a single Image Map image with 3 possible link destinations. Image Map image size needs to be: **300px W X 200px H at 72dpi**. Name this file: ***lastnameEx3ImgMap.psd***.
- b) 1 link needs to link to a page **inside** your site. This is called a **Relative** link. This needs to be a clearly indicated text-only link to a page inside your site.
- c) 2 of the links need to link **outside** of your site to content that is online. These are called **Absolute** links. You can use images (logos) or type for these links. Make sure they clearly communicate to the user where the link goes.
- d) The links on the image need to be organic and not laid-out in a grid. An example is provided for you on the Canvas site. Use this as a guide. **Yours needs to be completely different**. Your links need to communicate to the user **exactly** where they will go and should not be abstract.
- e) Save this file inside your **assets folder** which is inside your *lastnameEx3* folder.
- f) Save an **optimized version** of your image as **ex3ImageMap.png** (you can also save it as a **.gif** or **.jpg** depending on your design) in the *images directory* inside the *contact directory* of your sites root directory.
- g) From the **Files panel**, open the *contact.htm* page. Insert your cursor to the right of the **email image** and hit the **Return key** to create a paragraph <p> break. Insert the optimized **ex3ImageMap** image and **click on the image to select it**.
- h) In the **Properties Inspector panel**, you will see the word **Map** in the lower left corner with an input field to the right of it. Under this you will see a **Pointer Hotspot** and 3 shape-based image map/hotspot tools (*Rectangle*, *Circle* and *Polygon*). Select the appropriate hotspot tool for your link. You will probably use the polygon tool for all of them since they should not be laid-out in a grid.



- i) With the Image Map image selected, create a hotspot on top of your first link. Click + drag for the **Circle** and **Rectangle tool**, click + click with the **Polygon tool** until you have outlined the targeted area. Use the **Pointer Hotspot Tool** to select or deselect your hotspot.
- j) To create an **Absolute link** (off-site), select the hotspot you created with the **Pointer Hotspot Tool**. In the **Properties Inspector panel**, you should see a hashtag symbol (#) in the **Link field**. This is a placeholder, also known as a 'trigger'.
- k) Copy and paste the needed link into this **Link field replacing the hashtag**. Absolute links need to open a new browser window, not open in the same browser window where you linked from. To do this, use the **pull-down menu** next to the **Target** field under the **Link field**. Select **\_blank** as the target for the link. This opens the link in a new browser window when the user clicks it. **Do this with both Absolute links**.
- l) To create a **Relative link**, select the hotspot you created. Next to the **Link field**, click the folder icon and **navigate to a page inside your site**. Leave the **Target** field with **no target** (default). By default, all links will open in the same window as the clicked link.
- m) **File → Save All**. Preview the page in a browser. Preview the entire site using your grade sheet as a check list to make sure it's all working and follows the brief.
- You're done! Good job!

#### Requirements:

- Make sure *all image assets and onsite images are in the proper folders or directories and that everything is organized*.
- Make sure to *use the EXACT names as instructed in the brief*.
- Make sure to use *proper camelCase formatting*. No folder, directory or file should start with a capital letter. *Use capital letters on the webpage and on the images only*.
- You need a **new site map for the exercise**. Name it: **lastnameEx3SiteMap.pdf**. Hand this in inside the *assets* folder inside the **lastnameEx3** folder.
- Place your **assets folder**, your **root directory (onsite)** and your **site map (lastnameEx3SiteMap.pdf)** inside the folder **lastnameEx3**.
- Test everything in a browser! Keep the root clean!

#### **Handing in the assignment:**

- Compress your **lastnameEx3** folder (including the lastnameEx3SiteMap.pdf) into a .zip file named: **lastnameEx3.zip** and upload the .zip file via the Submit button on the **Exercise 3 page** in our Canvas course before the deadline. **Your upload may take a while to finish. Take this into account if you are uploading the assignment on the day it's due. Start your upload well before the posted deadline time so that it is not late!**